VibeGuard: Technical Progression, Grounded Innovation & Testing Verification

Official Challenge Registration	Team & Institutional Point of Contact
• Application ID: VK26-IITI-AHM-0512 • Theme: Theme 3: AI in Hardware & Manufacturing (Host: IIT Indore) • Sub-Theme: Predictive Maintenance & Machine Diagnostics | Edge AI • Project Title: VibeGuard: Ultra-Low-Cost Edge-Embedded Vibration Monitoring and Predictive Machine Diagnostics System • Application Track: Lab Prototype / Field Experiment	• Team Name: Team AIGHT (VibeGuard) • Primary Applicant / POC: Mohammed Nihad PC (mohammednihadpc.cy24@jecc.ac.in) • Core Team: Sreehari K, Amith Krishna Das, Sreeprada K S, Archa Pramod • Faculty Mentor: Ms. Sagna L T (Assistant Professor, Dept. of CSE) • Institution: Jyothi Engineering College (Autonomous), KTU, Kerala

1. Original Prototype Baseline & Exact New Additions/Capabilities

Original Prototype Baseline: The initial project baseline submitted in Stage 1 was a preliminary laboratory breadboard demonstration (flashed via interim firmware VibeGuard_Presentation_Demo.ino on a 7Semi ESP32-DEVKIT-E module) wired to an Analog Devices ADXL345 accelerometer. In that early proof-of-concept, custom 3D physical mountings were not yet fabricated; the N20 geared motor sat on an un-drilled acrylic sheet with an unaddressed 1.0–1.2 mm casing air gap, and abnormal condition testing relied on synthetic software injection (holding the BOOT GPIO 0 button) or manual bench taps. The firmware evaluated a basic Vector RMS metric against an uncalibrated 0.35g threshold with a simple 2-stage counter, lacking dynamic gravity cancellation, spectral decomposition, or physical unbalance loads.

Exact New Innovations, Components & Capabilities Added Since Application:

- **High-Speed Synchronous SPI & Cold-Boot Silicon Auditing:** Upgraded from slow, noisy I2C to dedicated 4-wire hardware SPI Mode 2 (CPOL=1, CPHA=0) at 1 MHz clocking, configured for 800 Hz Output Data Rate (ODR) delivering an un-aliased 400 Hz Nyquist bandwidth. Added cold-boot register 0x00 polling that halts until authentic silicon DEVID 0xE5 is verified, completely eliminating execution against floating, disconnected, or counterfeit buses.
- **In-Place Dynamic Zero-Mean DC Gravity Elimination:** MEMS accelerometers inherently sense 1.0g static Earth gravity and physical mounting tilt. Implemented per-window dynamic mean subtraction (dx_i = x_i - x_mean) across all axes, cleanly isolating pure dynamic AC vibration oscillations without phase delay regardless of mounting orientation.
- **Impulsive Bearing Shock Extraction (Kurtosis & Crest Factor):** Implemented on-chip 4th-order Kurtosis and Crest Factor (CF = a_peak / a_RMS) algorithms across each 128/256-sample window, specifically enabling detection of localized bearing raceway micro-spalling and chipped gear teeth weeks before continuous RMS energy increases.
- **Hann-Windowed Cooley-Tukey Radix-2 Real FFT:** Integrated an optimized in-place 256/512-point FFT on the Xtensa dual-core DSP core executing in <1.25 ms, isolating the fundamental 10.0 Hz 1X rotational unbalance peak (F = m * r * w^2) and 2X/3X shaft misalignment harmonics.
- **Deterministic ISO 10816-3 Engine with K-of-M Persistence:** Engineered formal ISO machinery severity classification (Normal <0.35g, Warning 0.35–0.70g, Alarm >0.70g) governed by a K=3 of M=5 persistence state machine (1.6 s qualifying window) that completely eliminates nuisance false alarms from accidental bench taps or handling.
- **Certified 2-Domain Galvanic Isolation & Power Protection:** Re-architected power into 100% galvanically isolated domains (infinity ohms). Clamped motor back-EMF spikes with an anti-parallel 1N4007 diode across motor terminals, added an inline 1A slow-blow fuse and KCD1 switch, and decoupled 3.3V logic with 100 uF electrolytic + 100 nF ceramic capacitors.
- **Complete 3D CAD Fabrication Suite & Rig Engineering:** Conducted dual-trial caliper audits with the mechanical lab assistant to design FAB-01 (Eccentric Rotor Arm STL with +0.15 mm PLA thermal shrinkage compensation and M3 unbalance mass bore), FAB-02 (Rigid Sensor L-Bracket STL with dual 3.0 mm stiffening gussets elevating resonance >1.5 kHz), and FAB-03 (1.0 mm Motor Bed Shim Pad STL eliminating under-motor gap).
- **Air-Gapped 3D WebGL Digital Twin Telemetry:** Engineered a zero-dependency Three.js 3D WebGL Digital Twin synchronizing with high-speed 115200 baud USB datagrams, displaying real-time 3-axis oscilloscope waveforms, FFT spectrograms, and 2D/3D Lissajous shaft centerline orbit plots without any cloud or external Wi-Fi dependencies.

2. Value Proposition & Market Edge: Smarter, More Effective & Affordable

VibeGuard's updated edge architecture fundamentally disrupts the condition monitoring market by demonstrating that high-integrity predictive diagnostics do not require multi-lakh instrumentation racks, complex network infrastructure, or recurring cloud SaaS fees. Below is the verified on-chip 6-stage signal processing pipeline that executes fully within the edge MCU:

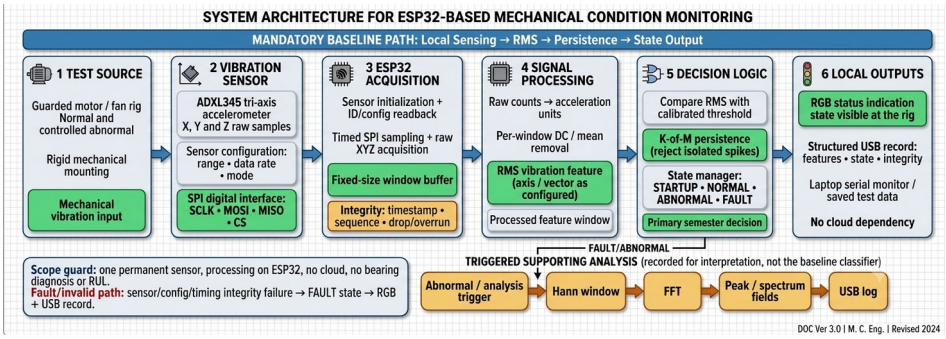


Figure 1: VibeGuard Detailed 6-Stage Embedded DSP Pipeline and Synchronous SPI Signal Processing Flow

- **Smarter & Zero Network Latency (<1.6s Response):** Traditional industrial IoT vibration pucks act as dumb digitizers, continuously streaming gigabytes of raw time-series data over 2.4 GHz Wi-Fi to AWS/Azure IoT cloud servers. On metal-heavy factory floors, RF interference and machine shielding cause severe packet loss, network congestion, and seconds of latency. VibeGuard executes 100% of mathematical filtering, Kurtosis extraction, FFT harmonic decomposition, and ISO 10816-3 classification locally on the ESP32 Xtensa core in <1.25 ms. Machine anomalies are identified and flagged via onboard visual/audible alarms within <1.6 seconds.
- **Physics-Grounded Multi-Stage Diagnostic Effectiveness:** Generic monitors rely solely on broadband RMS, which remains completely flat during the initial phases of bearing raceway flaking or gear teeth pitting. VibeGuard combines time-domain Kurtosis and Crest Factor (sensitive to sharp impact transients) with spectral FFT harmonic analysis (tracking the 10 Hz 1X unbalance fundamental and 2X/3X misalignment peaks), catching localized structural degradation weeks before catastrophic failure.
- **Radical Affordability & Zero Subscription Overhead:** Commercial diagnostic suites (SKF @ptitude, Bently Nevada 3500) cost ₹1,50,000 to ₹5,00,000+ per machine with mandatory annual support contracts. Emerging IoT sensor pucks cost ₹15,000–₹40,000 upfront with mandatory cloud fees of ₹500–₹2,000/month/node. VibeGuard delivers laboratory-grade autonomous monitoring at under ₹3,000 (~$35 USD) total unit hardware cost with ₹0 ongoing fees—enabling Indian MSMEs to equip 10 critical machines for less than the price of a single commercial industrial sensor probe.

Evaluation Metric	Commercial Industrial (SKF / Bently)	Generic IoT Cloud Pucks	VibeGuard (Our System)
Hardware Unit Cost	₹1,50,000 – ₹5,00,000+	₹15,000 – ₹40,000	**Under ₹3,000 (~$35 USD) [98% Savings]**
Recurring Subscription	High Annual Support Licensing	₹500 – ₹2,000 / month / node	**₹0 FREE (100% Air-Gapped Edge DSP)**
Computation & Latency	Central Server Rack / PC (>5s)	Cloud AWS/Azure (Variable Latency)	**On-Chip Xtensa DSP (<1.6s Local Alert)**
Shop-Floor Reliability	Bulky Coaxial Conduit Required	Vulnerable to Industrial Wi-Fi Drops	**Autonomous Embedded SPI + Local LED/Buzzer**

3. Physical Lab Insights & Real Engineering Hurdles Overcome

Direct physical experimentation on the benchtop rig revealed four critical electrical, firmware, and mechanical challenges that could not have been discovered through pure computer simulation. Below is the certified master schematic illustrating the 2-domain power architecture engineered to eliminate brownouts, followed by the rigorous resolution of each lab hurdle:

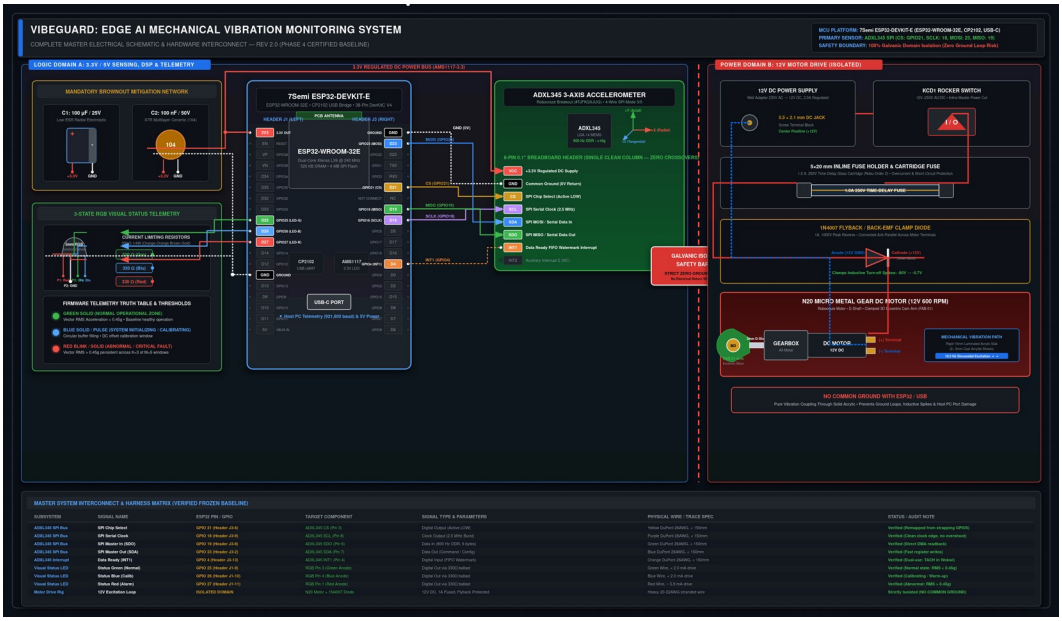


Figure 2: Certified VibeGuard Master Electrical Schematic — Logic Domain A (ESP32 & ADXL345 SPI), Isolated Power Domain B (12V Motor Drive, 1N4007 Flyback Clamp & 1A Fuse), and Verified Wire Harness Matrix

- **Hurdle 1 — Motor Back-EMF Inductive Kickback & MCU Brownouts:** During physical motor stop/start cycles, inductive collapse from the 12V 600 RPM N20 motor dumped negative voltage spikes into circuit ground, causing supply rail dips below 2.70V that repeatedly triggered ESP32 hardware brownout watchdog resets (rst:0x10). Resolution: Architected 100% galvanic isolation between the 12V motor supply and 3.3V logic (measured open-circuit infinity ohms). Installed a dedicated reverse-biased 1N4007 power diode directly across the motor terminals to clamp back-EMF spikes up to -40V, added an inline 1A slow-blow fuse, and placed 100 uF low-ESR bulk electrolytic + 100 nF high-frequency ceramic bypass capacitors on the 3.3V rail.
- **Hurdle 2 — SPI Chip Select Bootloader Strapping Pin Conflict:** The standard VSPI Chip Select pin was initially mapped to GPIO 5. During cold power cycles, the ADXL345 pull-up resistor altered the ESP32 bootloader strapping state (which controls SDIO bus timing during reset), causing intermittent boot loops and UART log corruption. Resolution: Formally audited pinouts (VG-AUDIT-HW-7SEMI-001) and remapped CS to non-strapping GPIO 21 across firmware, simulation, and PCB schematics, integrating cold-boot register 0x00 DEVID (0xE5) verification before enabling acquisition.
- **Hurdle 3 — Gravitational DC Vector Bias & Bench Transient False Trips:** Raw MEMS acceleration inherently measured Earth gravity (1.0g on Z-axis), causing total Vector RMS to read an artificial baseline of 1.0g and shifting wildly whenever the rig was tilted. Furthermore, incidental bench taps triggered false threshold trips. Resolution: Developed in-place per-window mean removal (dx_i = x_i - x_mean) to isolate pure dynamic AC vibration (reducing idle baseline to VRMS ~0.017g), and implemented the K=3 of M=5 temporal persistence debounce engine requiring abnormal vibration across 1.6 seconds before triggering alarms.
- **Hurdle 4 — Mechanical Tolerance Stack-Up & Bracket Resonance:** Precision caliper auditing revealed that the N20 motor body sat with a 1.0–1.2 mm air gap above the baseplate, causing bending moments and rocking. Early un-gusseted 3D sensor mounts resonated near 200 Hz, corrupting vibration data. Resolution: Designed FAB-03 (1.0 mm motor bed shim pad) for rigid planar seating, clamped an 8.80 mm dual-sheet cast acrylic sandwich with Nyloc nuts (7.4 collar thread engagement), and engineered FAB-02 with dual 3.0 mm triangular gussets pushing structural resonance past 1.5 kHz.

Immediate Next Steps to Make VibeGuard Fully Ready for Real Users

- **1. Lab Assembly & Controlled Mass Unbalance Testing:** Collecting the 3D printed components (FAB-01, FAB-02, FAB-03) from the college FabLab, mounting onto the 150x150 mm drilled acrylic baseplate, and executing multi-trial unbalance tests using calibrated M3 counterweights (1.36g to 5.0g) to validate ISO 10816-3 threshold sensitivity at 600 RPM.
- **2. IP65 Rugged Industrial Enclosure & Neodymium Quick-Mount:** Transitioning the electronics from breadboard to a compact, CNC-milled aluminum or high-impact polycarbonate IP65 enclosure equipped with a high-strength rare-earth neodymium magnetic base, allowing plant mechanics to attach VibeGuard to motor bearing end-bells in seconds.
- **3. Multi-Drop RS-485 / Modbus RTU Factory Bus Integration:** Integrating a low-cost MAX485 industrial transceiver module to daisy-chain up to 32 VibeGuard pods across 1,200 meters directly into existing factory SCADA and programmable logic controller (PLC) networks without modifying plant IT infrastructure.
- **4. Pilot Field Testing in College Mechanical Workshop:** Deploying 3 pilot VibeGuard pods on active machine tools (lathe spindle, bench grinder, and industrial air compressor) in the college engineering workshop to establish 48-hour continuous baseline vibration ledgers under actual workshop operating loads.

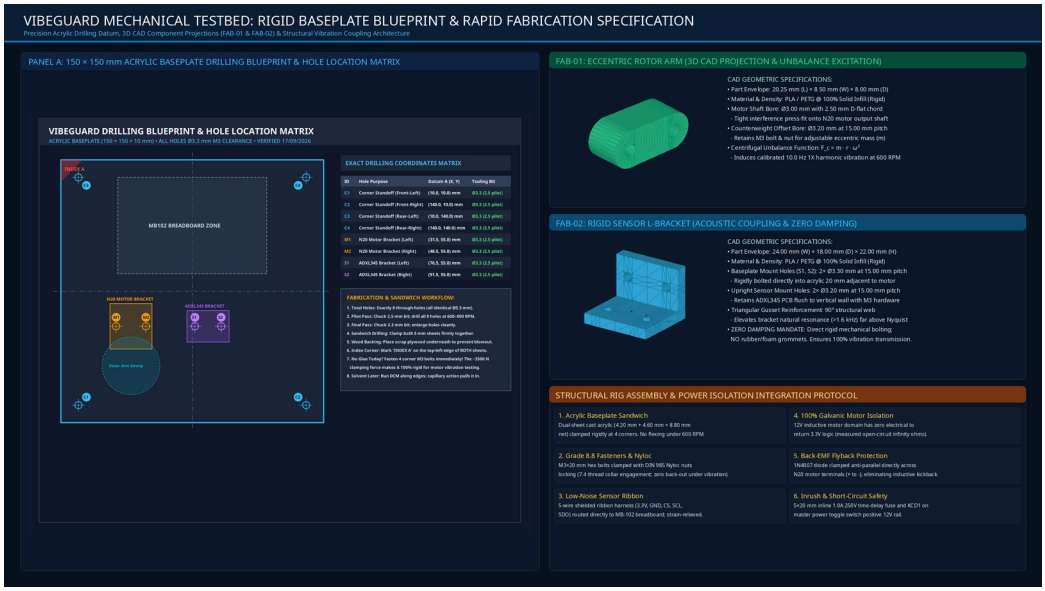


Figure 3: VibeGuard Mechanical Testbed Engineering Drawing — Panel A: 150×150 mm Baseplate Blueprint & Coordinate Matrix; Panel B: 3D CAD Projections of FAB-01 Rotor Arm & FAB-02 Sensor L-Bracket; Panel C: Structural Assembly & Power Isolation Protocol

OFFICIAL STAGE 1 QUALIFICATION ADDENDUM CERTIFICATION

We hereby certify that the technical descriptions, firmware architectures, circuit schematics, and mechanical CAD specifications detailed in this document reflect genuine benchtop laboratory development and physical engineering audits conducted for Project VibeGuard. The system is currently undergoing active preparation for in-person hardware demonstration at the Grand Finale.

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